Amir Kasaei

Research Assistant, RIML Lab & M.Sc. Computer Engineering, Sharif University of Technology

amirkasaei.com
Google Scholar
amirkasaei
a.kasaei@me.com
amirkasaei

EDUCATION

- **M.Sc. in Computer Engineering**
  Sharif University of Technology - (Sep 2023 – Oct 2026)
  Focus: Compositional Generation in T2I Diffusion Models
  Supervisor: Dr. M H Rohban
  GPA: 18.54/20

- **B.Sc. in Computer Engineering**
  University of Guilan (Sep 2019 – Aug 2023)
  Focus: Medical Image Analysis
  Supervisor: Dr. S A Mirroshandel
  GPA: 19.58/20 - *Ranked 1st*

RESEARCH INTERESTS

Computer Vision, Text-to-Image Generation, Diffusion Models, Compositional Learning, Multimodal Learning, Visual Reasoning, Vision–Language Alignment.

PUBLICATIONS

- **CARINOX: Inference-time Scaling with Category-Aware Reward-based Initial Noise Optimization and Exploration**
  Seyed Amir Kasaei, Ali Aghayari, Arash Marioriyad, Niki Sepasian, Shayan Baghayi Nejad, MohammadAmin Fazli, Mahdieh Soleymani Baghshah, Mohammad Hossein Rohban
  TMLR 2026
  arXiv:2509.17458

- **Evaluating the Evaluators: Metrics for Compositional Text-to-Image Generation**
  Seyed Amir Kasaei, Ali Aghayari, Arash Marioriyad, Niki Sepasian, MohammadAmin Fazli, Mahdieh Soleymani Baghshah, Mohammad Hossein Rohban
  NeurIPS 2025 Workshop GenProCC
  arXiv:2509.21227

- **Hidden Meanings in Plain Sight: RebusBench for Evaluating Cognitive Visual Reasoning**
  Seyed Amir Kasaei, Arash Marioriyad, Mahbod Khaleti, MohammadAmin Fazli, Mahdieh Soleymani Baghshah, Mohammad Hossein Rohban
  ICLR 2026 Workshop: From Human Cognition to AI Reasoning (HCAIR)
  arXiv:2604.01764

- **Erasure or Erosion? Evaluating Compositional Degradation in Unlearned Text-To-Image Diffusion Models**
  Arian Komaei Koma, Seyed Amir Kasaei, Ali Aghayari, AmirMahdi Sadeghzadeh, Mohammad Hossein Rohban
  CVPR 2026 Workshop on Machine Unlearning for Computer Vision
  arXiv:2604.04575

- **Weeding Out Bad Seeds: Initial-Noise-Robust Unlearning for Text-to-Image Diffusion Models**
  Arian Komaei Koma, Seyed Amir Kasaei, Aida Aryafar, Matin Ghiasi, Ali Aghayari, Amirhossein Souri, Mohammad Mosayyebi, AmirMahdi Sadeghzadeh, Mohammad Hossein Rohban
  Under Review at ICLR 2027
  arXiv:2609.37537

- **Erased but Exploitable: Black-box Embedding-Aware Prompting Against Unlearned Text-to-Image Diffusion Models**
  Arian Komaei Koma, Seyed Amir Kasaei, AmirMahdi Sadeghzadeh, Mohammad Hossein Rohban
  Under Review at TMLR
  arXiv:2605.26332

EXPERIENCE

Research Assistantships

- *RIML Lab, Sharif University of Technology* — Sep 2023 – Present
  Supervisor: Dr. Mohammad Hossein Rohban
  Focus: Compositional Generation, T2I Diffusion Models, Visual Reasoning, Multimodal Models

- *University of Guilan* — Sep 2022 – Aug 2023
  Supervisor: Dr. Seyed Abolghasem Mirroshandel
  Focus: Medical Image Analysis

Job Experience

- *ML Researcher, Matnafzar Houshmand* Nov 2025 – Present
  Developing a tool-based agentic visual question-answering pipeline for multimodal understanding, evaluation, and multi-step visual reasoning with large vision-language models (LVLMs).

Teaching Assistantships (Sharif University of Technology)

- *Data Mining* (Head TA) — Instructor: Dr. Mohammad Amin Fazli Sep 2025 – Jan 2026
  Duties: Led course coordination, quiz design, and grading; assisted with lectures, assignments, and student support.
- *System 2 in AI* — Instructors: Dr. Mohammad Hossein Rohban, Dr. Mahdieh Soleymani Baghshah Feb 2025 – Jun 2025
  Topics: Compositional Generation, Inference-time scaling, Test-time adaptation | *Duties:* Material prep and grading.
- *Deep Learning* — Instructor: Dr. Mahdieh Soleymani Baghshah Feb 2025 – Jun 2025
  Topics: Generative Models, VAE, GAN, Diffusion | *Duties:* Quiz and exam design, and grading.
- *Intelligent Analysis of Biomedical Images* — Instructor: Dr. Mohammad Hossein Rohban Sep 2024 – Jan 2025
  Topics: Diffusion and Generative Models | *Duties:* Supported exercises, exam/quiz design, and grading.

NOTABLE SELECTED COURSES

Master of Science:
Data Science in Software Engineering, Intelligent Analysis of Biomedical Images, Deep Learning, Algorithmic Game Theory, Software Engineering.

Bachelor of Science:
Fundamentals of Data Mining, Introduction to Deep Learning, Introduction to Machine Learning, Fundamentals of Language and Speech Processing, Software Testing.

HONORS & AWARDS

- **Exceptional Talent Admission**, Sharif University of Technology Sep 2023
  Awarded direct admission to M.Sc. in Computer Engineering as an exceptional talented student.
- **Ranked 1st**, B.Sc. Computer Engineering, University of Guilan 2019 – 2023
  Highest GPA among all undergraduate Computer Engineering students.

SKILLS

- **Machine Learning & Computer Vision:** Generative Models, Multi-modals, Diffusion, VLMs, Visual Reasoning
- **Programming Languages & Frameworks:** Python, PyTorch, TensorFlow, Keras, Java
- **Data Science & Developer Tools:** NumPy, Pandas, Matplotlib, W&B, Git
- **Soft Skills & Academic Leadership:** Teamwork, Leadership, Technical Communication, Mentorship, Problem Solving, Collaboration, Time Management

LICENSES & CERTIFICATIONS

- **IELTS** - Certificate
  IDP - Dec 2025, Overall: 7.5, Listening: 9, Reading: 7.0, Writing: 6.5, Speaking: 7.0

REFERENCES

Dr. Mohammad Hossein Rohban *Sharif University of Technology*
rohban@sharif.edu *Google Scholar*
Associate Professor of Computer Engineering, Sharif University of Technology, Tehran, Iran

Dr. Mohammad Amin Fazli *Sharif University of Technology*
fazli@sharif.edu *Google Scholar*
Associate Professor of Computer Engineering, Sharif University of Technology, Tehran, Iran

Dr. Mahdieh Soleymani Baghshah *Sharif University of Technology*
soleymani@sharif.edu *Google Scholar*
Associate Professor of Computer Engineering, Sharif University of Technology, Tehran, Iran